Name___________________

Complex Numbers and Polar Form

Date________________ Period____

Find the absolute value.

1) $4 + 3i$

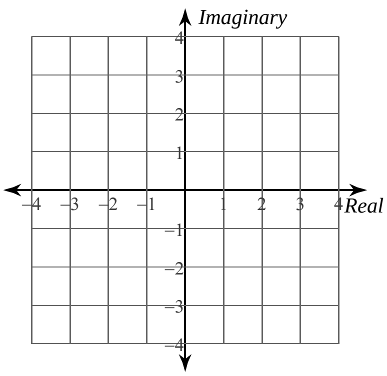
2) $-\sqrt{15} + i\sqrt{15}$

3) $3\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$

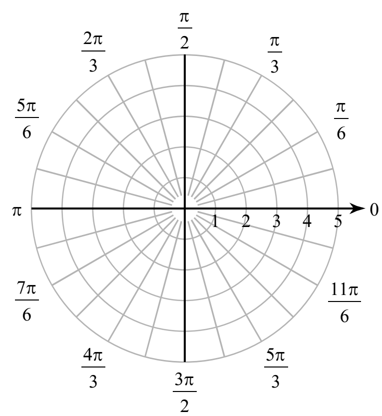
4) $\sqrt{21}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

Plot each point in the complex plane. Use rectangular coordinates when the number is given in rectangular form and polar coordinates when polar form is used.

5) $1 + 3i$



6) $3\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$



Convert numbers in rectangular form to polar form and polar form to rectangular form.

7) $-2\sqrt{3} - 2i$

8) $\sqrt{21} + i\sqrt{7}$

9) $-\frac{5\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i$

10) $2i$

11) $\sqrt{6}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

12) $2\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$